Engineering Safer, Cleaner, and More Efficient Combustion Environments

About Gaia Science

Gaia Science Pte Ltd is a Singapore-based scientific and engineering solutions provider serving research, healthcare and industrial sectors across Southeast Asia.

At the SSCI Combustion Meeting, we are showcasing our Strobic High Plume Exhaust Systems and advanced scrubber technologies, designed to manage hazardous exhaust, industrial emissions and combustion by-products while supporting environmental safety, regulatory compliance and energy efficiency.

With strong engineering expertise and a commitment to innovation, Gaia Science delivers practical solutions for safer, cleaner and more efficient operations.

Visit our booth to meet our specialists, discuss your exhaust and emissions-control challenges, and discover solutions tailored to your facility's needs. We look forward to connecting with you!





Industrial & Laboratory
Strobic Fans



High-Performance Exhaust Solutions
for Demanding Industrial and Laboratory Applications

Effective ventilation and exhaust are essential for maintaining a safe, clean and controlled working environment. Industrial and laboratory processes can generate corrosive fumes, chemical vapours, toxic gases, heat, dust and contaminated air that must be safely removed from the working area.

Strobic Fans are designed to provide reliable, high-volume exhaust airflow for demanding applications where conventional fans may not provide the required performance, durability or chemical resistance.

With robust construction and application-specific configurations, Strobic fans are suitable for laboratory exhaust systems, chemical processing, industrial ventilation and fume-control applications.

Key Benefits

High-volume exhaust performance
Provides reliable airflow for demanding ventilation applications.

Designed for contaminated air
Suitable for handling laboratory and industrial exhaust streams.

Corrosion-resistant construction
Available with materials and coatings selected for corrosive environments.

Reliable continuous operation
Designed for applications requiring dependable exhaust performance.

Low-maintenance design
Fan configuration helps simplify inspection and maintenance.

Flexible installation
Suitable for a wide range of laboratory and industrial exhaust systems.

Improved workplace safety
Helps remove hazardous fumes, vapours and contaminated air from occupied areas.

Typical Applications

Laboratory Exhaust

- Fume hood exhaust
- Chemical laboratory ventilation
- Research laboratories
- Pharmaceutical laboratories
- Biological and analytical laboratories
- Laboratory exhaust systems



Industrial Applications

- Chemical processing
- Semiconductor manufacturing
- Pharmaceutical production
- Battery manufacturing
- Metal finishing and treatment
- Process exhaust
- Corrosive gas exhaust
- Industrial ventilation



Why Choose a Strobic Fan?

Strobic fans are particularly suitable for applications where the exhaust air may contain corrosive chemicals, fumes, vapours or other contaminants.

The fan can be integrated with a complete exhaust treatment system, such as:



This configuration allows contaminated air to be captured, treated where required, and safely discharged from the facility.

Application-Specific Solutions

Fan selection should be based on the required airflow, static pressure, operating temperature, contaminant characteristics, corrosion requirements and installation environment.

Our team can assist customers in selecting the appropriate fan configuration and supporting equipment for their application.

Complete Exhaust System Solutions

We can support customers from application assessment and fan selection through installation, commissioning, troubleshooting and preventive maintenance.

Strobic Fan + Scrubber + Exhaust System

A reliable solution for laboratory, industrial and process exhaust applications.



Gaia Science

Authorized Distributor

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Industrial & Laboratory
Scrubber Systems

Effective Gas Treatment Solutions for Safer, Cleaner and Compliant Operations

Industrial and laboratory processes can generate hazardous, corrosive, toxic, odorous or environmentally harmful gases and vapours. Propor exhaust gas treatment is essential to protect personnel, equipment and the environment while supporting regulatory and workplace safety requirements.

Our Industrial and Laboratory Scrubber Systems are designed to effectively remove or reduce unwanted contaminants from exhaust gas streams before they are released into the environment.



Key Benefits

Effective pollutant removal
Reduces hazardous gases, vapours, fumes and odours.

Enhanced workplace safety
Protecs personnel from exposure to harmful chemical contaminants.

Corrosion protection
Reduces corrosive contaminants before they reach downstream systems.

Environmental protection
Minimizes the release of harmful substances into the atmosphere.

Application-specific design
Scrubber configuration and materials can be selected according to your process, gas composition, and operating conditions.

Suitable for laboratory and industrial applications
From small-scale laboratory and materials to large-industrial process exhaust systems.

Types of Scrubber Systems



Wet Scrubbers
Use a liquid scrubbing solution to absorb, neutralize, or react with contaminants. Suitable for many inorganic gases, acidic gases, alkaline gases and water-soluble contaminants.



Packed Bed Scrubbers
Provide a large contact area between the exhaust gas and scrubbing liquid, making them suitable for continuous gas absorption and chemical neutralization.



Venturi Scrubbers
Use high-velocity gas flow and liquid injection to achieve intensive gas-liquid contact. Suitable for high gas flow and efficiency and particular removal.



Dry Scrubbers
Use solid sorbent or reactive media to capture specific gases and vapours without generating liquid wastewater.



Activated Carbon Scrubbers
Use activated carbon media to absorb VOCs, solvents, odours and selected chemical vapours.

Typical Applications

Laboratory Exhaust
- Fume hood exhaust
- Chemical laboratory ventilation
- Research laboratories
- Pharmaceutical laboratories
- Biological and analytical laboratories
- Laboratory exhaust systems

Industrial Applications
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- Metal finishing and treatment
- Process exhaust
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- Industrial ventilation

Application-Based Selection

The right scrubber should be selected based on:

- Type and concentration of contaminant
- Exhaust flow rate
- Moisture content
- Regulating efficiency
- Operating conditions

Our team can assist customers in selecting the appropriate fan configuration and supporting equipment for their application.

Our End-to-End Support

- Application and maintenance & selection
- Installation & commissioning
- Maintenance & servicing

From Process Exhaust to Clean Air



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